Dibya Bhusal

Founder & Technical Lead · Full-Stack, Backend & IoT Engineer

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SUMMARY

Technical lead at MIMO, where I run the Raspberry Pi print kiosks, payments and Firebase backend that students use daily. Strongest in TypeScript/Node.js, Python, Firebase and React.

EDUCATION

REVA University Aug 2024 – May 2028

B.Tech, Computer Science & Engineering (IoT, Cybersecurity & Blockchain) Bengaluru, India

Coursework: Data Structures & Algorithms, Software Engineering, Cybersecurity, IoT, Blockchain, Artificial Intelligence

EXPERIENCE

Technical Lead — MIMO (Vision Printt Technologies) Sep 2026 – Present

Campus print kiosks (Raspberry Pi + CUPS, Firebase, UPI), 2 in production; joined as IoT Hardware Intern. Bengaluru

printmimo.tech

- Lead engineering on a 10-contributor production codebase (444 of my commits): set up CI/CD deploy gates, secret scanning, coverage checks, CODEOWNERS and the contributing guide; 534 backend tests.
- Traced production kiosk lag to an unbounded download pipeline (125 concurrent threads, swap 900/990 MiB on a 1 GB Raspberry Pi) and replaced it with a bounded worker pool, stale-job filtering and cache eviction, covered by 15 new regression tests.
- Fixed a race that showed customers another customer's file by allocating print codes in atomic Firestore transactions and serialising print dispatch on the Pi to one job at a time.
- Stopped false refunds from slow CUPS responses (lpstat timeout + retry, real failure reasons logged) and raised API capacity from 10 to 50 instances for a ~7,000 requests/hour lunch peak.
- Made refunds exactly-once across 3 entry points with a shared atomic claim, and made the Cashfree payment webhook signature-verified and idempotent.
- Resolved production incidents (403 upload failures, 12 lost secrets, broken Office conversion), closed an auth bypass, locked down public Firestore/Storage rules and fixed 13 audit findings.

PROJECTS

Project 117 — On-Premise Multi-Agent Industrial AI | *Python, FastAPI, Ollama, LanceDB, Next.js, Kotlin* Live · GitHub

- Built solo a refinery digital twin (58 equipment, 224 sensors) where 3 local LLM agents diagnose a fault, plan a reroute and verify safety through 19 permission-gated tools; nothing executes unless the validator and safety agent both agree.
- Kept plant data on-premise with clearance-filtered retrieval (5 levels), default-deny egress and a hash-linked, Ed25519-signed audit trail; shipped a Next.js console and an offline Android field app.

NeighborHealth — Dengue Forecasting for Bengaluru | *Python, OCR, FastAPI, Next.js, GitHub Actions* Live · GitHub

- Turned 606 government PDF bulletins into a weekly case series (374/379 reconcile exactly) and forecast with 6 models, rolling backtests and conformal intervals; the 1-week ensemble beats seasonal-naive by 13.6% (MAE 208 vs 241).

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C/C++, Kotlin, Move

Backend & Cloud: Node.js/Express, FastAPI, Firebase (Firestore, Cloud Functions, Auth), Cloud Run, PostgreSQL, Redis, Docker, GitHub Actions

Frontend: React, Next.js, Vite, Tailwind CSS, Three.js / React Three Fiber

AI / ML: Ollama (local LLMs), multi-agent systems, LangGraph, RAG (LanceDB, BM25, pgvector), scikit-learn, forecasting

Systems & Security: Linux, Raspberry Pi, CUPS, SSH, ESP32, MQTT, RBAC, Firestore rules, webhook verification

ACHIEVEMENTS

- Hackathons: 1st place and Solana track winner, TRI-ZEROTRO (Solana/Anchor anchoring pipeline, team of 3); 4th place, Namma Security 2.0 national hackathon, KAVACH (automated CCTV, fraud-scoring and FIR pipeline).
- State-level athlete: cricket, representing the state team in U16 & U19 national tournaments; table tennis, state senior team.